NITHIN DEEPA

United States | (270) 917-4048 | deepanithinroy9@gmail.com | LinkedIn | GitHub | Portfolio

EDUCATION

Murray State University **Aug 2023 - May 2027**

B.S., Computer Science (GPA: 3.4)

- **Coursework:** Data Structures, Algorithms, Machine Learning, Artificial Intelligence, Database Systems, Software Development

TECHNICAL SKILLS

AI & Generative AI: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), LangChain, LangGraph, Prompt Engineering, Embeddings, Vector Search
Programming Languages: Python, Java, JavaScript, TypeScript, SQL, C++
Frameworks & Libraries: FastAPI, Flask, React, Node.js, REST APIs, Server-Sent Events (SSE)
Databases: PostgreSQL, SQL Server, ChromaDB, pgvector
Cloud, DevOps & Developer Tools: AWS, Google Cloud (Cloud Run), Docker, Git, GitHub, Render, Google Cloud Vertex AI

WORK EXPERIENCE

Padsnap (AI Startup) | *AI Engineer Intern* **May 2026 - Jul 2026**

- Built a retrieval-augmented generation (RAG) pipeline with LangChain, ChromaDB, and Hugging Face embeddings for semantic retrieval over real estate data, improving answer relevance and grounding responses across 500+ interactions.
- Designed an LLM-powered content generation pipeline that automated property descriptions, listing headlines, social media captions, and email marketing content.
- Implemented real-time LLM response streaming through three Flask API endpoints using Server-Sent Events (SSE) and the Groq streaming API.
- Orchestrated multi-step AI agent workflows in LangGraph by combining retrieval, tool calls, and task automation to deliver source-cited answers over pgvector.

Sodexo — Student Lead (Promoted from Student Employee) **Nov 2023 - Jul 2026**

- Collaborated with 10+ team members to maintain standard operating procedures (SOPs) and training documentation, supporting operational consistency and compliance.

PROJECTS

AI Code Review Assistant | Python, FastAPI, LangChain, RAG, GitHub API, PostgreSQL, Docker

- Built an AI agent that reviews GitHub pull requests using RAG over 10k+ lines of code, flagging defects with file- and line-level citations and catching 75% of planted bugs on a 50-PR test set.
- Added AI-suggested tests, human approval before posting comments, and automated eval tests, which cut manual review time by 40%.

OpsPilot — Enterprise AI Operations Copilot | Python, FastAPI, React, TypeScript, LangGraph, pgvector

- Engineered a full-stack AI operations platform enabling document intelligence, conversational querying, and automated task workflows.
- Implemented a RAG pipeline with document parsing, embeddings, and PostgreSQL/pgvector vector search to deliver source-grounded answers.

AI Outreach Agent | Python, FastAPI, React, TypeScript, PostgreSQL, Gemini API, Vercel, Render

- Built an AI outreach platform using Gemini for call screening and message generation, with human approval before any message is sent.
- Secured it with login, per-user data isolation, consent checks, rate limiting, and duplicate-message protection, covered by 30+ automated tests.

LEADERSHIP EXPERIENCE

Indian Students Association - Murray State | *President* **Feb 2024 - Feb 2026**

- Led a 200+ member student organization, overseeing budget management, event planning, and cross-cultural initiatives to promote student engagement and campus inclusion.
- Organized 10+ campus-wide events**,** coordinating activities and connecting international students with academic resources, professional development opportunities, and university support services